

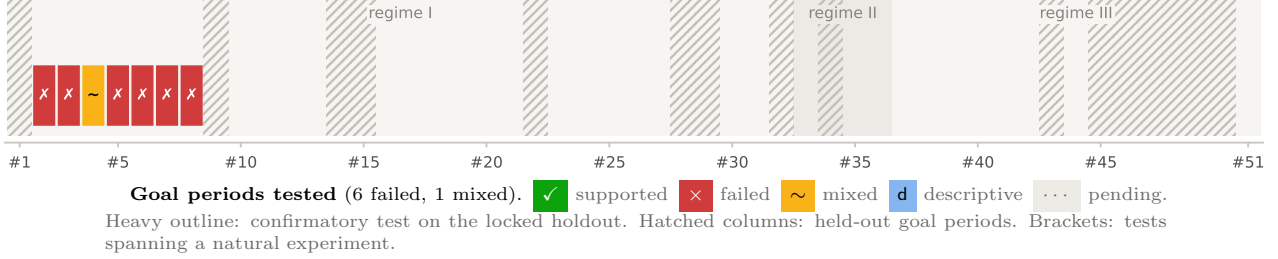
H124: Four agents for weeks: an exact kinetic Ising benchmark for the mean-field approximations (#4, #6)

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 2/5 Usefulness 2/5

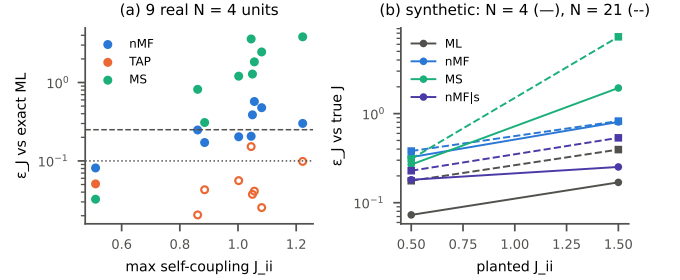
Bottom line. Mean-field inversions of the kinetic Ising match exact likelihood only when agents' own states do not persist; on the village's call clock they mostly do, so use exact ML.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

With four agents the kinetic Ising likelihood is exact and cheap, so I compared exact maximum likelihood (ML) with three mean-field inversions (naive MF, TAP, Mézard–Sakellariou MS) on nine closed-roster regime-I units, on the per-call talk clock and a 1-min grid. Synthetic worlds on the real #4/#6 and #51 schedules came first. The HH's claim fails: TAP or MS is within 10% of ML in 1 of 9 units. That unit (4c) is the only one with weak own-state persistence (self-coupling $J_{ii} \leq 0.6$). Elsewhere (J_{ii} 0.86–1.22) TAP has no solution on 25–75% of rows, MS overshoots the couplings up to $\times 3.8$, and naive MF shrinks them (median error 0.25). The synthetic shows the same law at $N = 4$ and $N = 21$. Off-diagonal couplings are at noise level in 7/9 units. Holdout: not run.



Observables. (a) Off-diagonal coupling error of each inversion against exact ML, per real unit, against the unit's largest self-coupling (open circles: TAP solved on only some rows; dotted 10%, dashed 25%). Only the weak-persistence unit is inside 10% for all three. (b) Synthetic error against the true J (medians; $N = 4$ on the 4c schedule at $\sigma_J = 0.3$, $N = 21$ on #51's, pooled over σ_J): MS and naive MF break as J_{ii} grows; exact ML and the own-spin stratified nMF (A1) hold.

PHYSICAL MODEL

Model 02, kinetic Ising. At each call of agent i , its talk spin $s'_i = \pm 1$ (did the call produce chat) depends on the four latest talk spins, its own included. J_{ij} : coupling of i to j 's state; J_{ii} : persistence of i 's own state; h_i : field. From i 's rows: $m'_i = \langle s'_i \rangle$, C the covariance of $\mathbf{s}$, D_i the covariance of s'_i with $\mathbf{s}$. The inversions differ only in the gain a_i ; TAP has a real solution only when $\sum_j B_{ij}^2 (1 - m_j'^2) \leq \frac{4}{27} (1 - m_i'^2)$, $B_i = D_i C^{-1}$.

$$P(s'_i = +1 | \mathbf{s}) = [1 + e^{-2(h_i + \sum_j J_{ij} s_j)}]^{-1} \quad (\text{exact ML: logistic})$$

$$J_i = a_i^{-1} D_i C^{-1}, \quad a_i^{\text{nMF}} = 1 - m_i'^2$$

$$a_i^{\text{TAP}} = (1 - m_i'^2) [1 - (1 - m_i'^2) \sum_j J_{ij}^2 (1 - m_j'^2)]$$

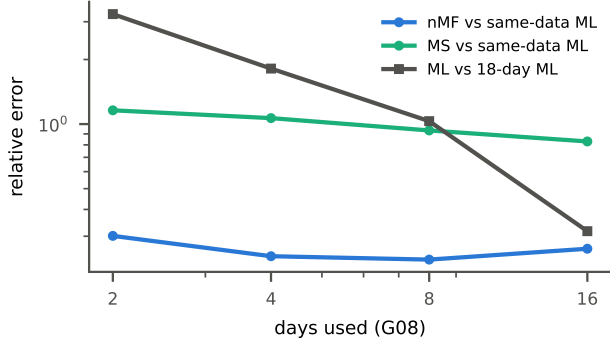
$$a_i^{\text{MS}} = \langle 1 - \tanh^2(g_i + x \sqrt{\Delta_i}) \rangle_x, \quad \Delta_i = J_i C J_i^T$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

A method rule for this project, with a number. On the call clock, agents' own states persist ($J_{ii} \approx 0.9$ –1.2), and then TAP and MS are not usable and naive MF underestimates couplings by 5–55%. Exact per-row logistic ML costs seconds at $N = 21$, so large- N kinetic Ising fits here should use it (or pseudo-likelihood) and report J_{ii} . Mean-field inversions are acceptable on channels without persistence (1-min talk: MS within 10% in 6/9 units). For an operator nothing changes; this protects the coupling estimates other hypotheses report.

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 (HH) TAP or MS within 10% (G04/G06)	1/4 units (4c: TAP 0.05, MS 0.03)	failed
Kill: none within 25%	2/4 primary units (threshold); 4/9 all	at threshold
P2 naive MF > 10%	8/9; shrink 0.45–0.95	supported
P3 MS ≤ TAP < nMF	only 4c; nMF best full-cover in 8/9	failed
P4 ranking carries to $N = 21$	synthetic, best method agrees 15/16 cells	supported
P5 off-diagonal J near null	7/9 inside shift null; ML noise 0.6–1.1	supported
P6 1-min activity nMF > 25%	7/8 units	supported
N1 coupling lower in #6 than #4	no difference ($\bar{J}$ 0.004–0.05)	failed
N2 ML noise below bias by 8 days	bias flat; crossover 16–18 days	mixed



Second figure. Native N2 on G08 (18 days, 20 subsamples per length). The inversions’ bias against same-data ML is flat in the number of days, while ML’s distance to the full 18-day fit falls from 3.3 to 0.32. Below about 16 days, sampling noise dominates any inversion’s bias.

GOAL PERIODS IN DETAIL

G02 × failed
G03 × failed
G04 ~ mixed
G05 × failed
G06 × failed
G07 × failed
G08 × failed

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`witeup/paper.tex`).

CAVEATS

The benchmark compares inversions with ML on the same data, and the off-diagonal ML couplings are themselves noise in 7/9 units, so coupling errors partly measure noise handling. The EP error ε_σ is not interpretable (reference EP at noise level). Plefka2[t] (Aguilera 2021) was not implemented. The own-spin stratified variants (A1) were added after the synthetic, before real data. Units with 2–3 days have degenerate intervals.

ROUND 2

Repeat the benchmark where couplings are real (regime-III named talk, H50) and add Plefka2[t] once its notes are in the literature folder.

- **Essence.** which coupling estimator to trust at $N = 21$; round 1 says exact (or pseudo-) likelihood on the call clock, because own-state persistence breaks the mean-field inversions.
- **H124-R1** Large-N rule.
- **H124-R2** Plefka2[t].
- **H124-R3** Real couplings.
- **H124-R4** Markov order.

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